

STATISTICS FOR DATA ANALYTICS

APPLICATION ASSIGNMENT

50-Row E-Commerce / Retail Dataset

1. DATA UNDERSTANDING :

- Identify all numerical variables:
Age, Order_Value, Items_Purchased, Discount_Percent, Delivery_Days, Customer_Rating, Marketing_Spend
- Identify all categorical variables.
Customer_ID, Gender, Repeat_Customer, Payment_Mode, Order_Value_Band
- Identify which numerical variables are discrete.
Age, Items_Purchased, Discount_Percent, Delivery_Days, Customer_Rating, Order_Value
- Identify which numerical variables are continuous.
Marketing_Spend
- Check for missing values.
Missing Values Not Presented
- Check for duplicate Customer_ID values.
Duplicate Customer_ID Not Appeared
- Identify the variable that represents customer spending.
Order_Value
- Identify the variable that represents customer satisfaction.
Customer_Rating
- Identify the variable that represents delivery performance.
Delivery_Days
- Identify the variable that represents marketing investment.
Marketing_Spend

2. MEAN, MEDIAN & MODE :

- Use Order_Value to calculate the mean.
MEAN IS 6939 Rs
- Use Order_Value to calculate the median.
MEDIAN IS 5500 Rs
- Use Order_Value to find the mode, if applicable.
MODE IS Not Applicable
- Compare mean and median.
Mean Is Greater Than Median
- Which measure represents a typical customer order value better?
Median – 5500 Rs
- Explain why.
Median is less affected by the very high order values.
- Business question: What is the typical customer spend?
Typical customer spend is around 5500 Rs because the median represents the middle order value.

3. VARIANCE & STANDARD DEVIATION :

- Use Order_Value to calculate the variance.
VARIANCE IS 29088229
- Calculate standard deviation.
STANDARD DEVIATION IS 5393.35 Rs
- Is customer spending tightly concentrated or widely spread?
Customer spending is Widely Spread.
- What does the standard deviation tell you?
Customer spending shows considerable variation compared with the mean value.
- Why is standard deviation easier to communicate than variance?
Because standard deviation is in the same unit as Order_Value, while variance is in squared units.
- Does a high standard deviation necessarily mean the data is bad?
No
- Business question: If one store has a higher standard deviation, what does it mean?
It means the store has higher variation in customer spending.

4. QUARTILES, PERCENTILES & IQR :

- Calculate Q1.
Q1 IS 3587.50 Rs
- Calculate Q2.
Q2 IS 5500 Rs
- Calculate Q3.
Q3 IS 8325 Rs
- Calculate IQR.
IQR IS 4737.50 Rs
- Calculate P25, P50, P75 and P90.
P25 IS 3587.50 Rs
P50 IS 5500 Rs
P75 IS 8325 Rs
P90 IS 12650 Rs
- What does Q1 mean?
Approximately 25% of customers have Order_Value below 3587.50 Rs.
- What does Q3 mean?
Approximately 75% of customers have Order_Value below 8325 Rs.
- What percentage of observations lie below Q3?
75% of observations lie below Q3.
- What does P90 mean?
Approximately 90% of customers have Order_Value below 12650 Rs.
- What does IQR tell us about customer spending?
IQR shows the spread of the middle 50% of customer spending.
- Business challenge: Which percentile would you use to identify the top 10% customers?
I would use P90.

5. OUTLIER INVESTIGATION :

- Calculate Q1, Q3 and IQR.
Q1 IS 3587.50 Rs
Q3 IS 8325 Rs
IQR IS 4737.50 Rs

- Calculate Lower Boundary and Upper Boundary.
LOWER BOUNDARY IS -3518.75 Rs
UPPER BOUNDARY IS 15431.25 Rs
- Identify potential outliers.
15500 Rs, 18000 Rs, 22000 Rs and 30000 Rs
- Are these observations necessarily errors?
No
- Give possible business reasons for a very high order.
Bulk Purchase, Premium Products, Business Customers, Special Events or Multiple Products Purchased Together.
- Calculate the mean before investigating the outlier.
MEAN BEFORE OUTLIER IS 6939 Rs
- Recalculate the mean after excluding potential outliers.
MEAN AFTER OUTLIER IS 5683.70 Rs
- How much did the mean change?
Mean decreased by 1255.30 Rs.
- Should a Data Analyst automatically delete an outlier?
No. First investigate whether it is a genuine transaction or a data-entry error.

6. HISTOGRAM & BOXPLOT :

- What does the shape look like?
Right-skewed
- Where are most orders concentrated?
Most orders are concentrated in the lower-to-middle spending range.
- Is the distribution symmetric?
No
- Does it appear right-skewed or left-skewed?
It appears right-skewed.
- Are there extreme values?
Yes. Some customers spend much more than most customers, especially above 15000 Rs.
- Identify the median.
Median is 5500 Rs.
- Identify Q1 and Q3.
Q1 is 3587.50 Rs and Q3 is 8325 Rs.
- Identify the IQR.
IQR is 4737.50 Rs.
- Identify potential outliers.
15500 Rs, 18000 Rs, 22000 Rs and 30000 Rs.
- Business question: Which visualization is better for identifying outliers?
I would use a Boxplot because it clearly shows potential outliers separately from the main data.

7. SKEWNESS & KURTOSIS AWARENESS :

- Is Order_Value symmetric, right-skewed or left-skewed?
The Order_Value distribution is Right-Skewed.
- Explain what the skewness tells you about customer spending.
Most customers have lower or medium spending, while a few customers have very high spending.

- Check the kurtosis using Python/Excel.
KURTOSIS IS approximately 6.393
- Does the distribution show heavier or lighter tails compared with a normal distribution?
The distribution shows relatively heavier tails compared with a normal distribution.
- Business interpretation.
Most customers make small or medium-value purchases, while a few customers make very high-value purchases. This creates a right-skewed distribution.

8. PROBABILITY :

- Probability that a randomly selected customer is a Repeat Customer.
PROBABILITY IS 0.32 OR 32%
- Interpretation.
There is a 32% probability that a randomly selected customer is a Repeat Customer.
- Probability that an order has a discount greater than 20%.
PROBABILITY IS 0.28 OR 28%
- Interpretation.
There is a 28% probability that a randomly selected order has a discount greater than 20%.
- Probability that an order takes more than 5 days to deliver.
PROBABILITY IS 0.10 OR 10%
- Interpretation.
There is a 10% probability that a randomly selected order takes more than 5 days to deliver.

9. CONDITIONAL PROBABILITY :

- Repeat Customer given Order_Value > ₹5,000.
PROBABILITY IS 46.43%
- Calculate the overall probability of being a Repeat Customer.
OVERALL PROBABILITY IS 32%
- Compare the two probabilities.
Conditional Probability is 46.43% and Overall Probability is 32%.
- Which probability is higher?
Conditional Probability is higher.
- What does the difference suggest?
Customers with Order_Value greater than ₹5,000 have a higher chance of being Repeat Customers.
- Does this prove that high spending causes repeat purchases?
No
- Give one possible factor other than Order_Value that could influence repeat purchasing.
Customer satisfaction or product quality can influence repeat purchasing.

10. RANDOM VARIABLES :

- Identify two discrete random variables from the dataset.
Items_Purchased and Delivery_Days
- Identify two continuous random variables from the dataset.
Order_Value and Marketing_Spend
- Explain why you classified each variable that way.
Items_Purchased and Delivery_Days are countable whole-number values. Order_Value and Marketing_Spend can conceptually take different monetary values.

- Define $X = \text{Order_Value}$ of the selected customer. Explain what possible values X can take.
 X represents the Order_Value of a randomly selected customer. X can take values present in the dataset, such as 850 Rs, 1200 Rs, 5500 Rs, 18000 Rs or 30000 Rs.
- Define $Y = \text{Items_Purchased}$. Explain how Y differs from X .
 Y represents the number of items purchased. Y is a countable value, while X represents the customer's Order_Value .

11. SAMPLING & SAMPLING BIAS :

- How would you select the sample?
I would randomly select 500 customers from the 50,000 customers.
- Would you use probability or non-probability sampling?
I would use Probability Sampling.
- Which sampling technique would you recommend?
I recommend Stratified Random Sampling.
- Why?
It gives different customer groups a fair chance of being represented and helps reduce sampling bias.
- What could go wrong if the sample is not representative?
The survey results may not correctly represent the overall customer population.
- Scenario 2: Is surveying only customers who purchased during the last 7 days necessarily representative?
No
- Who might be missing?
Customers who did not purchase during the last 7 days may be missing.
- What type of bias could occur?
Selection Bias could occur.
- How would you improve the sampling approach?
I would select customers from a wider time period and randomly sample from the complete customer population.
- Scenario 3: Identify the sampling approach and discuss its limitation.
This is Convenience Sampling. It selects customers who are easiest to contact, but it may not represent the overall customer population and can cause sampling bias.

12. NUMPY SAMPLING SIMULATION :

- Select a sample and calculate the average Order_Value .
I randomly selected 10 customers. One sample produced a mean of approximately 7790 Rs.
- Repeat the process 100 times. What are the results?
Average of sample means ≈ 6770.55 Rs
Standard deviation of sample means ≈ 1649.49 Rs
Sample means varied from approximately 3540 Rs to 11550 Rs.
- Are all sample means identical?
No. Each sample contains a different combination of customers, so the sample mean changes.
- What happens when sample size increases from 10 to 25?
The sample means become more stable because a larger sample represents the population better.
- Compare the standard deviation of sample means.
Sample size 10 SD ≈ 1649 Rs
Sample size 25 SD ≈ 793 Rs

13. NUMPY IMPLEMENTATION :

- Mean.
₹6,939
- Median.
₹5,500
- Variance.
₹29,088,229
- Standard Deviation.
₹5,393.35
- Minimum.
₹850
- Maximum.
₹30,000
- 25th Percentile.
₹3,587.50
- 50th Percentile.
₹5,500
- 75th Percentile.
₹8,325
- 90th Percentile.
₹12,650
- IQR.
₹4,737.50
- Note on variance and standard deviation.
These figures use the population formula (ddof=0). If the 50 rows are treated as a sample, sample variance is approximately ₹29,681,866.33 and sample standard deviation is approximately ₹5,448.11.

14. FINAL BUSINESS INTERPRETATION :

- Customer spending.
Mean is ₹6,939 and median is ₹5,500. High-value orders pull the average upward, so management should use the median when discussing typical customer spending.
- Customer spending distribution.
Order_Value is right-skewed. Most customers spend moderate amounts, while a small group spends significantly more.
- Outliers.
Four orders are potential outliers: C004, C007, C009 and C044. They may be genuine high-value transactions and should be investigated.
- Customer behaviour.
Repeat probability is 32% overall and 46.43% among customers spending more than ₹5,000. Higher spending and repeat purchasing are associated in this dataset.
- Sampling.
The company should use a representative probability-based sampling method, preferably stratified random sampling.

15. CORRELATION VS CAUSATION :

- Marketing_Spend vs Order_Value – Correlation.
 $r \approx 0.857$. This is a Strong Positive Correlation.

➤ Interpretation.

Customers/orders associated with higher marketing spend also tend to have higher order values.

➤ Can we say that increasing marketing spend causes Order_Value to increase?

No. Correlation does not prove causation. Other factors such as customer type, product category, season and promotions may be involved.

➤ Delivery_Days vs Customer_Rating – Correlation.

$r \approx -0.582$. This is a Moderate Negative Relationship.

➤ Interpretation.

As delivery days increase, customer ratings tend to decrease.

➤ Discount_Percent vs Order_Value – Correlation.

$r \approx -0.066$. This is a Very Weak Negative Relationship, effectively close to zero.

➤ How can causation be established?

A controlled experiment or randomized A/B test would provide stronger evidence for causation.

16. COVARIANCE :

➤ Calculate the sample covariance between Marketing_Spend and Order_Value.

COVARIANCE IS approximately 5,433,311.78

➤ What does the positive sign mean?

Marketing_Spend and Order_Value generally move in the same direction.

➤ What does the magnitude mean?

It shows the amount of joint variation, but it is difficult to interpret directly because covariance depends on the units.

➤ Why is covariance difficult to interpret?

Covariance is affected by the measurement scale.

➤ Why is correlation easier to interpret?

Correlation is standardized between -1 and +1, making it easier to understand and compare relationships.

➤ Business interpretation.

Higher marketing investment is associated with higher order values. However, the company cannot conclude that increasing marketing spend will definitely cause sales to increase.